

Week 2

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- solving non singular system
- solving singular system
- solving sys of equation with more var
- matrix row reduction
- Row operations that Preserve singularity
- Rank of matrix
- Row echelon form
- Reduce Gaussian elimination algo

① solving systems

1.1 non singular

$$\begin{array}{rcl} 2a + 5b & = & 46 \\ 8a + b & = & 32 \\ \times -5 & & \\ \hline -40a - 5b & = & -160 \\ + 2a + 5b & = & 46 \\ \hline -38a & = & -114 \\ \therefore a & = & 3 \\ \therefore b & = & 8 \end{array}$$

1.2 singular (inconsistent) (contradiction)

$$\begin{array}{rcl} a + b & = & 10 \\ 2a + 2b & = & 24 \\ \Rightarrow a + b & = & 10 \\ -a - b & = & -12 \\ \hline 0 & = & -2 \end{array}$$

no solution

1.3 singular (infinite solution)

$$\begin{array}{rcl} 5a + b & = & 11 \\ 10a + 2b & = & 22 \\ \therefore \text{equation 2} & = & \text{equation 1} \\ \text{so there are infinite} & & \\ \text{solution} & & \\ \text{and also system is singular} & & \end{array}$$

② matrix row reduction

ex $\begin{array}{l} 5a + b = 17 \\ 4a - 3b = 6 \end{array}$

$$\begin{array}{l} a + 0.2b = 3.4 \\ \Rightarrow 0a + b = 2 \end{array}$$

$$\begin{array}{l} b = 2 \\ \Rightarrow a = 3 \end{array}$$

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

it is called Row echelon form :
main diag is one and under this
diag everything is zero
main diag: bunch of one's followed
by perhaps bunch of zeros

notes

- Switching Rows, multiply Rows [with non zero scale], adding Rows to other row will keep singularity and non singularity, it just change det value

Rank of matrix: it is related to space that needed to store corresponding item

ex

note high Rank, high info

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

has one solution

$\therefore$ Dimension of solution space = 0

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

has infinite solution

on same line
dimension of solution space = 1

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

has infinite solution

on any point
 $\therefore$ dimension of solution space = 2

in general:

Rank = no of Piece of info

ex $a+b+c=0$
 $a+b+2c=0$
 $a+2b+c=0$

$\therefore$ 3 info Provided

$\therefore$ Rank = 3

ex $a+b+c=0$
 $a+b+2c=0$
 $a+b+3c=0$

here eq 3 = eq 1 + eq 2

$\therefore$ 2 info Provided

$\therefore$ Rank = 2

ex $a+b+c=0$
 $2a+2b+2c=0$
 $3a+3b+3c=0$

only 1 info Provided

$\therefore$ Rank = 1

Row echelon form

ex $\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \xrightarrow{\text{step 1}} R_1 \rightarrow \frac{R_1}{5} \text{ and } R_2 \rightarrow \frac{R_2}{4} \Rightarrow \begin{bmatrix} 1 & 0.2 \\ 1 & -0.75 \end{bmatrix}$

step 2

Subtract Row 1 From Row 2 $\Rightarrow R_2 \rightarrow R_2 - R_1 \Rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & -0.95 \end{bmatrix}$

step 3

$R_2 \rightarrow \frac{R_2}{-0.95} \Rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix} \Rightarrow$ Rank = no of 1's in diagonal
 $\therefore$ Rank = 2

and if no zeros in diag $\Rightarrow$ matrix is non singular

Pivot: First element in row equals non zero

Rank = no of Pivot

ex $\begin{bmatrix} 2 & 3 & 1 \\ 0 & 5 & 4 \\ 0 & 0 & 7 \end{bmatrix}$ Pivots are 2, 5, 7
note all numbers under Pivot should equal zero

here Rank = 3 (number of Pivot)

Gaussian elimination Algo

- std way to solve linear equation system or Find inverse of matrix by Convert matrix to Row echelon form.

operation example

① exchange Rows

② add Rows

③ substitution

ex $2a - b + c = 1$

$2a + 2b + 4c = -2$

$4a + b = -1$

$\begin{bmatrix} 2 & -1 & 1 \\ 2 & 2 & 4 \\ 4 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$

① Need to make Pivot = 2 on 1st Row to = 1

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$$\therefore R_1 = \frac{R_1}{2} \Rightarrow R_1 = \begin{bmatrix} 1 & -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 \end{bmatrix}$$

② Remove any number under 1 to be zero

$$\therefore R_2 = R_2 - 2 R_1 \rightarrow \text{new}$$

$$\therefore R_2 = \begin{bmatrix} 2 & 2 & 4 \end{bmatrix} - \begin{bmatrix} 2 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 3 \end{bmatrix} \begin{bmatrix} -3 \end{bmatrix}$$

and $R_3 = R_3 - 4 R_1 \rightarrow \text{new}$

$$\therefore R_3 = \begin{bmatrix} 4 & 10 & 0 \end{bmatrix} - \begin{bmatrix} 4 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 3 & -2 \end{bmatrix} \begin{bmatrix} -3 \end{bmatrix}$$

$\therefore$ new array

$$\begin{bmatrix} 1 & -0.5 & 0.5 \\ 0 & 3 & 3 \\ 0 & 3 & -2 \end{bmatrix} \begin{bmatrix} 0.5 \\ -3 \\ -3 \end{bmatrix}$$

to make 3 = 1 and any number under it = 0

$$\textcircled{1} R_2 = \frac{1}{3} R_2 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \end{bmatrix}$$

$$\textcircled{2} R_3 = R_3 - 3 R_2 = \begin{bmatrix} 0 & 3 & -2 \end{bmatrix} \begin{bmatrix} -3 \end{bmatrix} - 3 \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \end{bmatrix}$$

$$R_3 = \begin{bmatrix} 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix}$$

$\therefore$ new array =

$$\begin{bmatrix} 1 & -0.5 & 0.5 \\ 0 & 1 & 1 \\ 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} 0.5 \\ -1 \\ 0 \end{bmatrix}$$

$$R_3 = \frac{-1}{5} R_3 \Rightarrow \text{new array} = \begin{bmatrix} 1 & -0.5 & 0.5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ -1 \\ 0 \end{bmatrix}$$

$$R_2 = R_2 - R_3 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \end{bmatrix}$$

$$\Rightarrow \text{new array} = \begin{bmatrix} 1 & -0.5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ -1 \\ 0 \end{bmatrix}$$

$$R_1 = R_1 - \frac{1}{2} R_3 = \begin{bmatrix} 1 & -0.5 & 0 \end{bmatrix} \begin{bmatrix} 0.5 \end{bmatrix}$$

$$R_1 = R_1 - (-0.5 R_2) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix}$$

$\therefore$ Final array =

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

$$\therefore a = 0$$

$$b = -1$$

$$c = 0$$

in singular system

• by row echelon form we can find if system:

① has Infinite solution

② has no solution

ex

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & -7 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 9 \\ 0 \end{bmatrix}$$

has infinite solution

ex

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & -7 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 9 \\ 4 \end{bmatrix}$$

has no solution

Lab 1 week 2

Topics:

① use numpy linear algebra Package to Find solution to system

② evaluate determinant of matrix

③ explore numpy.linalg

① solving linear equation using matrices

$$\begin{cases} 4x_1 - 3x_2 + x_3 = -10 \\ 2x_1 + x_2 + 3x_3 = 0 \\ -x_1 + 2x_2 - 5x_3 = 17 \end{cases}$$

using numpy

```
A = np.array([
    [4, -3, 1],
    [2, 1, 3],
    [-1, 2, -5]])
```

```
b = np.array([-10, 0, 17]):
```

to find shape or size

```
np.shape(A) ; // (3, 3)
```

```
np.shape(b) ; // (3, )
```

to solve equation $\Rightarrow x = \text{np.linalg.solve}(A, b)$; // 011 [1. 4. -2.]

② evaluate determinant of matrix

```
d = np.linalg.det(A) ; // -60
```

note if you Try to solve singular equation, linalg will Throw LinAlgError because matrix is singular